Dataset Overview

1 Purpose of the Released Materials

This supplementary document provides an overview of the main derived datasets accompanying the study. Its purpose is to clarify the structure, origin, and intended use of the released resources, with particular attention to the person-level datasets used for baseline modeling.

The released materials do not include raw videos or directly identifying information. Instead, they consist of derived tabular representations obtained through privacy-preserving processing pipelines applied to annotated video material.

2 Released Dataset Branches

The supplementary materials include three main dataset branches:

- a monocular dataset derived from CAM1;
- a monocular dataset derived from CAM2;
- a multi-view 3D dataset derived from real triangulation and controlled fusion.

Each released dataset is organized as a CSV file in which each row corresponds to a person-level temporal window.

3 Common Labeling Scheme

All released datasets are based on the same reduced set of target labels used for baseline modeling:

- Stand
- Sit
- ClosedPosture
- DisengagedRepetitive

These labels were obtained from the full annotation codebook through a conservative aggregation step designed to reduce sparsity and improve statistical stability. The complete annotation scheme and the mapping from original labels to reduced labels are described in the accompanying `codebook_final.pdf` document.

4 Temporal Representation and Windowing

Across all dataset branches, the temporal representation follows the same person-level logic.

First, annotated intervals were aligned to person identities. Then, temporal timelines were discretized at 2 Hz, corresponding to one timestep every 0.5 seconds. Sliding windows were subsequently generated over these timelines using the same strategy adopted throughout the study.

The released datasets are centered on 2-second windows, which were selected because they provided the most stable baseline behavior in the monocular experiments.

5 CAM1 and CAM2 Datasets

The CAM1 and CAM2 datasets were constructed from 2D pose-based person-level features extracted independently from each monocular view. In both cases, the dataset construction pipeline followed the same main steps:

1. person-to-annotation identity matching;
2. timeline construction at 2 Hz;
3. sliding-window generation;
4. extraction of interpretable aggregated features.

The released CSV files correspond to the final window-level datasets used for baseline experiments on CAM1 and CAM2.

6 Multi-view 3D Dataset

The multi-view 3D dataset was constructed from the real multi-view reconstruction pipeline described in the paper. In this branch, person-level windows were derived from frame-level 3D outputs produced by triangulation and controlled fusion.

The 3D branch followed the same overall methodological logic as the monocular branches, but replaced monocular 2D feature extraction with a multi-view 3D representation. The construction pipeline included:

1. frame-level 3D output generation;
2. identity attachment through the validated CAM1 mapping branch;
3. person-level timeline construction at 2 Hz;
4. sliding-window generation;
5. extraction of aggregated 3D features.

The final released 3D datasets correspond to the ML-ready and trainable subsets derived from this pipeline.

7 Feature Types

The released datasets contain derived numerical and metadata variables. Depending on the branch, these include:

- temporal metadata;
- person identifiers;
- window configuration variables;
- label variables;
- aggregated motion and posture descriptors;
- multi-view quality and coverage descriptors.

In the multi-view 3D branch, the released features mainly describe the availability, density, and reliability of the 3D signal within each temporal window, rather than fine-grained joint-wise kinematics.

A complete description of all released variables is provided in `data_dictionary.csv`.

8 Final Released Files

The supplementary package includes the following dataset files:

- CAM1 person-level dataset;
- CAM2 person-level dataset;
- multi-view 3D ML-ready dataset;
- multi-view 3D trainable dataset.

These files represent the main derived dataset contribution accompanying the paper.

9 Privacy and Ethics

The released materials are limited to derived representations and do not include raw audiovisual data. No raw face imagery, voice recordings, or biometric identity models are distributed. The released datasets follow the same privacy-preserving and non-diagnostic principles that guided the overall project.